BUKOMA JUMA MOYA

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SOFTWARE ENGINEER

Backend-focused software engineer with strong experience designing scalable systems, real-time data pipelines, and distributed architectures. Proven ability to build production-grade systems using Node.js, Spring Boot, PostgreSQL, and cloud-native tooling. Experienced in microservices, event-driven systems, authentication, and performance optimisation.

TECHNICAL SKILLS

Frontend: React.js, HTML5, CSS3, Material UI, Bootstrap, Redux, Axios

Backend: Node.js, Express.js, Spring Boot, GraphQL, REST APIs, Firebase Functions, WebSockets, Socket.IO

Databases: PostgreSQL, MongoDB, MySQL, Oracle Database, Microsoft SQL Server, Redis, Firebase Firestore

Cloud / DevOps: Docker, Kubernetes, Git, Gitflow, GitHub Actions, Google Cloud Platform, Microsoft Azure, AWS

Security: JWT, OAuth2, OpenID Connect, RBAC, bcrypt, SSL/TLS, RFID verification, secure password storage

Testing / Tools: Jest, Mocha, Postman, Swagger/OpenAPI, Thunder Client, Chrome DevTools, Firebase Crashlytics, Cloud Logging

Protocols / Standards: MQTT, MQTT 5.0, IEC 61850 MMS/GOOSE, Modbus-TCP, OPC UA, DNP3-IP

PROFESSIONAL EXPERIENCE

Auri Energy Limited - R&D Software Developer

May 2025 – Present

Cetus Central (Enterprise IoT Platform)

- Designed a microservices-based architecture enabling real-time ingestion and processing of industrial IoT data across distributed systems
- Built low-latency data pipelines (Kafka + Redis + PostgreSQL) achieving sub-millisecond state reads

- Implemented event-driven APIs and streaming systems for high-frequency sensor data
- Integrated industrial communication protocols (MQTT, OPC UA, Modbus) for grid-compliant systems
- Developed and coordinated 6 core platform modules handling Power Quality, Fault Outage, Energy Integrity, Topology Twin, Communication Data and Metering Data.
- Enforced secure system design (JWT, RBAC, TLS) aligned with regulatory standards
- Built CI/CD pipelines with Docker & Azure, enabling zero-downtime deployments
- Built database schemas and migration scripts, and developed Jest/Mocha test suites that reached 90% code coverage.
- Achieved ~90% test coverage across unit and integration layers
- Collaborated in agile teams to translate requirements into scalable system features

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Whave Solutions Limited- System Architect & Planner

February 2025

Smart Water Dispenser System (IoT + FinTech Integration)

- Architected a scalable IoT + cloud system integrating ESP32 devices, RFID authentication, and mobile payments
- Designed secure API layer connecting IoT devices, admin dashboards, and mobile interfaces
- Modelled transaction-heavy PostgreSQL schema and recommended time-series storage for telemetry
- Conceptualised a React-based administrator dashboard and a mobile-friendly user portal for balance checks and top-ups.
- Defined end-to-end system flows (payments, credit systems, water usage tracking)
- Evaluated and planned mobile money integrations
- Designed fault-tolerant edge computing approach for offline-first environments

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Shamos Tech Solutions - Software Developer Intern

Nov 2023 – Feb 2025

SUDI App (Marketplace Platform – PERN Stack)

- Built RESTful APIs handling user management, listings, payments, and chat systems

- Designed and optimised PostgreSQL schemas, improving query efficiency and response times
- Implemented secure authentication (OAuth2 + JWT) across all endpoints
- Developed real-time features using WebSockets for messaging and notifications
- Applied caching strategies and performance tuning to handle high user activity
- Wrote unit and integration tests (Jest) ensuring system reliability
- Automated CI/CD pipelines, reducing deployment friction

KauntaBook (SME Financial Platform)

- Built backend services using Node.js + Firebase Functions (serverless architecture)
- Designed real-time financial systems for transactions, invoicing, and reporting
- Implemented secure authentication and data protection mechanisms
- Optimised performance using Firestore queries and caching strategies

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PROJECT EXPERIENCE

LearnEasy - Peer Learning Platform (MERN Stack)

Mar 2024 – Jun 2024

- Developed a fullstack platform supporting collaborative learning and real-time interaction
- Built role-based system (students, facilitators, admins) with RBAC
- Designed scalable backend with MongoDB, Redis caching, and service-layer architecture
- Implemented CI/CD pipelines and containerised deployment (Docker + Kubernetes)
- Applied secure coding practices (bcrypt, TLS, CSRF/XSS protection)
- Integrated real-time communication (Socket.IO)

ADDITIONAL EXPERIENCE

Kampala Capital City Authority (KCCA) - IT Intern

Jun 2023 – Aug 2023

- Contributed to system design and database normalisation for enterprise systems
- Built components of a queue management system (Java + Spring Boot)

- Managed Active Directory environments and network infrastructure
- Provided IT support and assisted in enterprise system research and analysis

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EDUCATION

Makerere University
Bachelor of Information Systems and Technology, Systems Development
2022 – 2024

King's College Budo
UACE (2019 – 2020)
UCE (2015 – 2018)

KEY STRENGTHS

- Strong foundation in distributed systems and real-time architectures
- Ability to move from system design to implementation, and then deployment
- Experience with high-performance backend systems and IoT integrations
- Deep understanding of security, authentication, and scalable APIs
- Comfortable working across backend, cloud, and DevOps layers

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REFERENCES

1. Mr. Nyaruhuma Christopher
   Chief Technology Officer
   Auri Energy Limited
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2. Nyakaisiki Monica Rose
   Deputy Director Information and Communication Technology
   Kampala Capital City Authority
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3. Kayondo Muhsin Musa
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   Shamos Tech Solutions
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